ResearchPaper Lens — Forensic & Analytical Report

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1. Document Integrity & Cryptographic Evidence

Metric / Digest	Value
Filename:	sample_03_edge_case_math.txt
File Size:	640 bytes
Pages / Words / Sentences:	1 pages | 87 words | 6 sentences
References Section:	None detected
SHA-256 Digest:	`820a772937f19562d2303c49dcb07275dff5125d52081947ab9b3af69c724427`
SHA-1 Digest:	`df66aef236285484f1b008a7930d73f802281196`
MD5 Digest:	`0845270ef0ebf4c875a6961eb2dbd880`

2. Explainable Review Cues & Limitations

- Extractive Faithfulness: TextRank extracts verbatim sentences directly from source text (0% hallucination risk).
- Graph Centrality: Vertex importance is determined via PageRank (d=0.85, max_iter=100) over TF-IDF cosine similarities.
- Chronological Ordering: Extracted sentences are strictly restored to their original document sequence.
- Scanned PDF Limitation: Low-text heuristic (< 100 chars/page) flags image-only PDFs where embedded text is minimal.
- Lexical Search Boundary: Passage retrieval uses lexical TF-IDF matching and does not perform neural semantic re-ranking.
- Privacy by Default: Sensitive personal identifiers, emails, and credentials are automatically masked.

3. TextRank Extractive Summary (38 words, 43.7% compression)

Extracted 2 sentences out of 6 using TF-IDF PageRank graph centrality.

Recent breakthroughs by Dr. Smith et al. in Information Retrieval (i.e. IR systems) have achieved state-of-the-art results on benchmark datasets (e.g. TREC, SQuAD). As illustrated in Fig. 1 and Eq. 2, the error rate dropped to 3.1415 percent.

4. Academic Baseline Summaries

Method	Output
Lead-k: *(First k sentences)*	Recent breakthroughs by Dr. Smith et al. in Information Retrieval (i.e. IR systems) have achieved state-of-the-art results on benchmark datasets (e.g. TREC, SQuAD). As illustrated in Fig. 1 and Eq. 2, the error rate dropped to 3.1415 percent.
Word-Frequency: *(Term count weighting)*	Recent breakthroughs by Dr. Smith et al. in Information Retrieval (i.e. IR systems) have achieved state-of-the-art results on benchmark datasets (e.g. TREC, SQuAD). Furthermore, Prof. Davis demonstrated that subsampling reduces convergence time significantly.

5. Salient Keywords (TF-IDF Aggregation)

dropped: 0.41	**error**: 0.41	**illustrated**: 0.41
fig: 0.41	**percent**: 0.41	**rate**: 0.41
alpha: 0.38	**fixed**: 0.38	**particular**: 0.38
parameter: 0.38		

Confidentiality & Privacy: Redactions applied (0 sensitive patterns masked). Document evaluated locally with ResearchPaper Lens.